

Regression Analysis

- What is Regression?
- Why Regression?
- Linear Regression
 - Linear Regression algorithm using least square method
 - Evaluation of method
- Multilinear Regression
- Logistic Regression

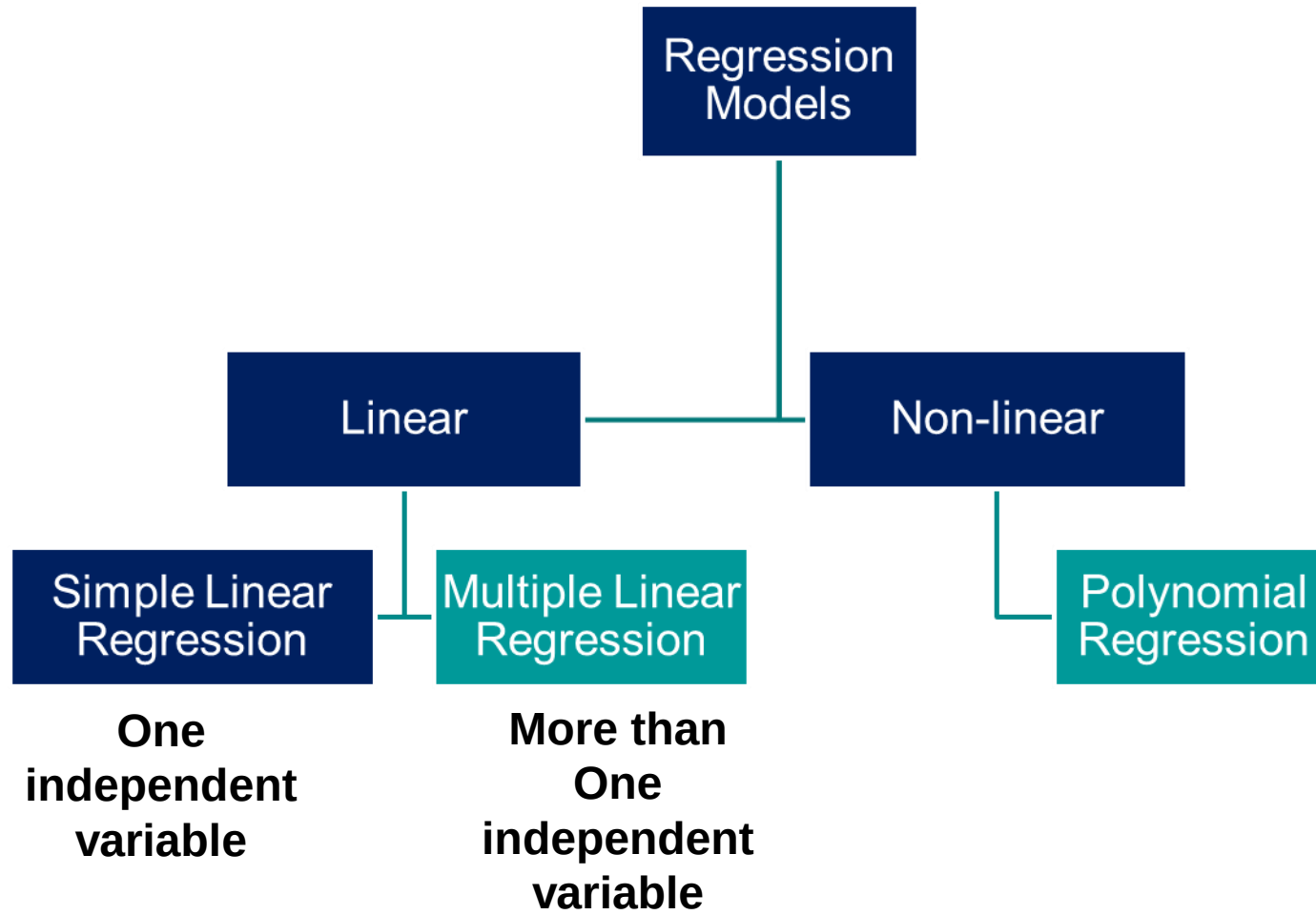
What is Regression?

- Regression is a **supervised learning algorithm** under Machine Learning terminology
- An important tool in **Predictive Analytics**
- Regression analysis is a predictive modeling technique which investigates the relationship between a dependent and independent variable.
- Graphing a line over a set of data points that most closely fits the overall shape of the data.
- The regression shows the changes in the dependent variable on the Y axis to the changes in the explanatory variable on X axis.

What is Regression?

- Regression is a tool for finding **existence of an association relationship** between a dependent variable (Y) and one or more independent variables ($X_1, X_2, \dots, X_n$) in a study.
- The relationship can be linear or non-linear.
- A dependent variable (**response variable**) “measures an outcome of a study (also called **outcome variable**)”.
- An independent variable (**explanatory variable**) “explains changes in a response variable”.

Types of Regression



Most Common Regression Algorithms

- Simple linear regression
- Multiple linear regression
- Polynomial regression
- Multivariate adaptive regression splines
- Logistic regression
- Maximum likelihood estimation (least squares)

Use cases of Regression

- Predictive analytics
 - Operation efficiency
 - Supporting decisions
 - Correcting errors
 - New insights
-
- House Price Predictions
 - Trend forecasting
 - E.g. what will be the price of gold in next six months
 - Finding Associations among attributes:
 - E.g. Mediclaim agencies: Effect of age on claims

Linear Regression

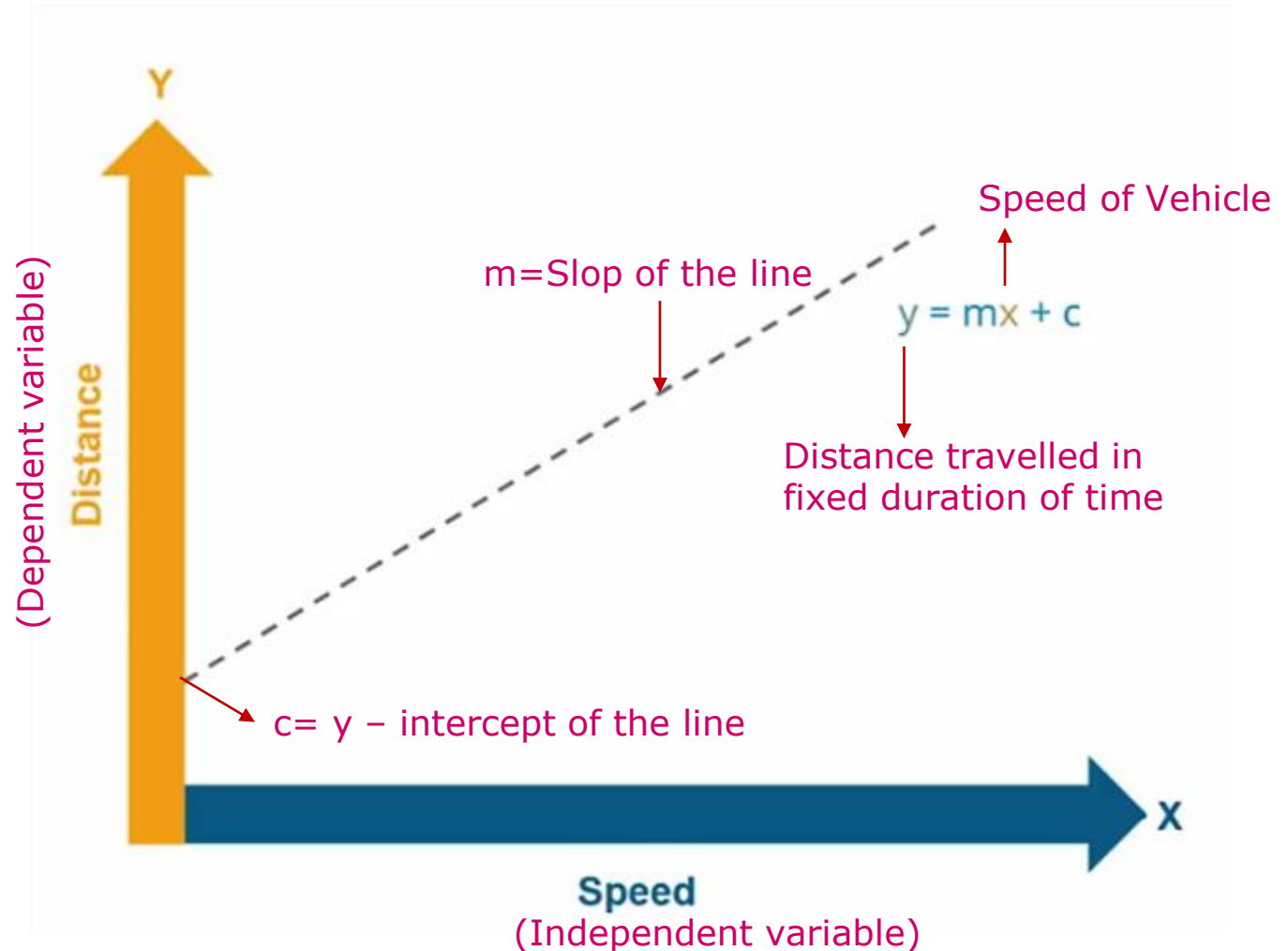
- Linear regression: It is a linear approach to modelling the relationship between a scalar response and one or more explanatory variables (also known as dependent and independent variables).
- The case of one explanatory variable is called **simple linear regression**; for more than one, the process is called **multiple linear regression**.
- In linear regression, the relationships are modeled using linear predictor functions whose unknown model parameters are estimated from the data.
- Linear regression models are often fitted using the least squares approach.

Simple Linear Regression

- One of the easiest algorithm in machine learning.
- Simple Linear regression: It is a statistical model that attempts to show the relationship between two variables through the linear equation.
- Data is modeled using a straight line ($Y = mX + c$)
- Correlation between X and Y variables

Simple Linear Regression: Understanding

+ve Relationship



Simple Linear Regression: Understanding

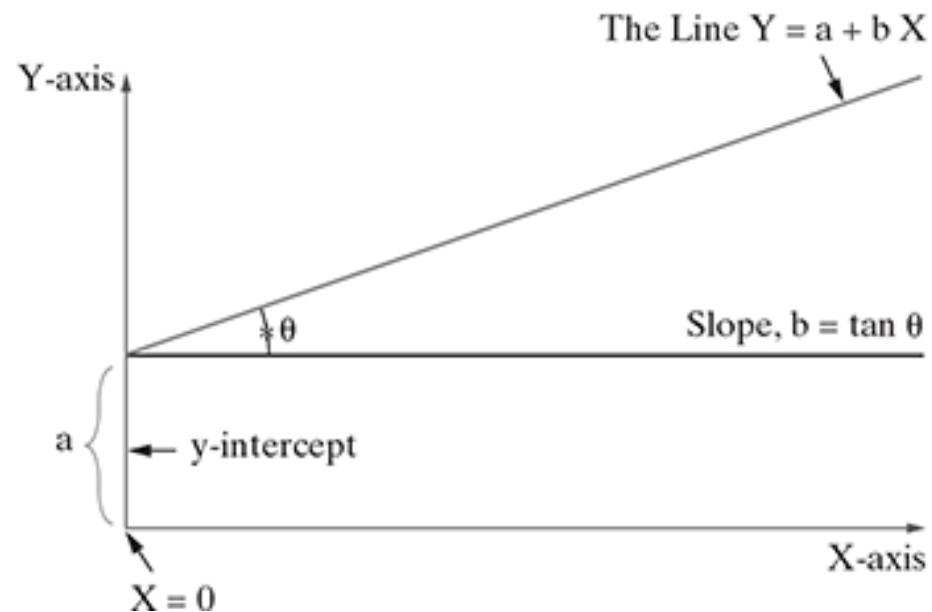
Dependent variable $\leftarrow$

$Y = a + b X$

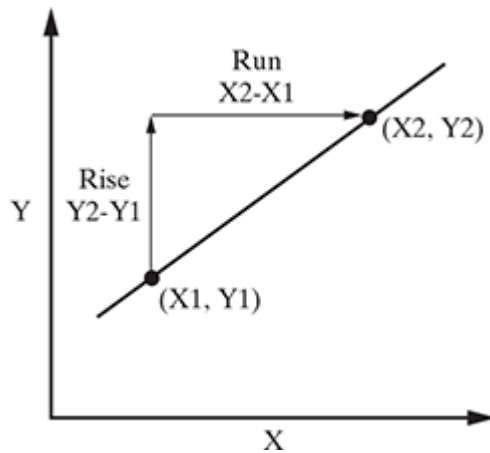
Independent variable $\rightarrow$

Y-intercept $\leftarrow$

Slope of the line $\rightarrow$



Slops of Simple Linear Regression Model



$$\text{Slope} = (Y_2 - Y_1) / (X_2 - X_1) = \Delta(Y) / \Delta(X)$$

Slope = Change in Y / Change in X

$$\text{Slope} = \frac{\text{Rise}}{\text{Run}} = \frac{Y_2 - Y_1}{X_2 - X_1}$$

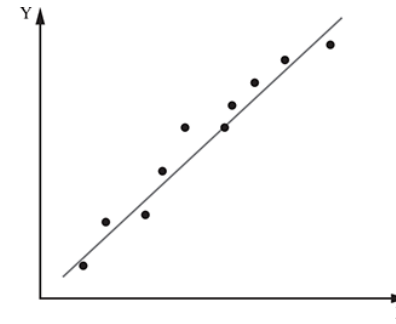
Example:

$(X_1, Y_1) = (-3, -2)$ and $(X_2, Y_2) = (2, 2)$

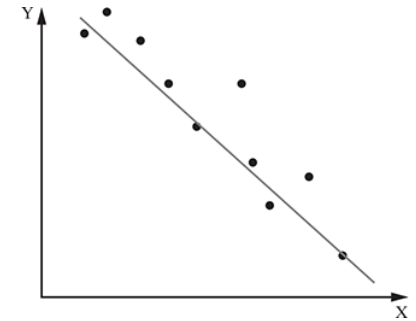
Rise = $(Y_2 - Y_1) = (2 - (-2)) = 2 + 2 = 4$

Run = $(X_2 - X_1) = (2 - (-3)) = 2 + 3 = 5$

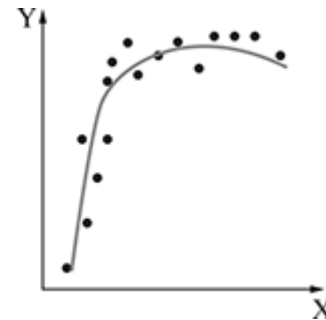
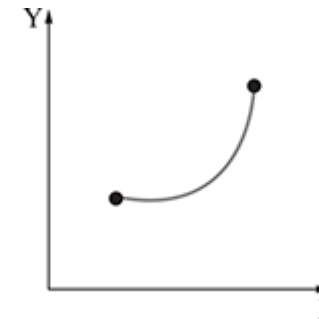
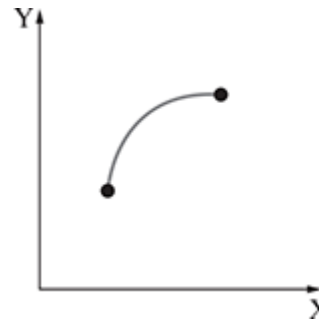
Slope = $\text{Rise} / \text{Run} = 4 / 5 = 0.8$



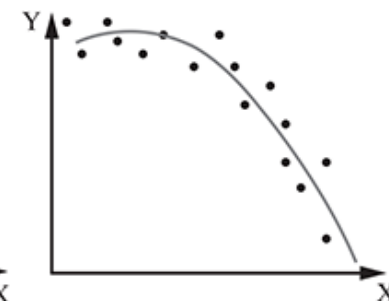
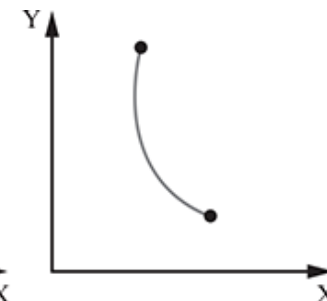
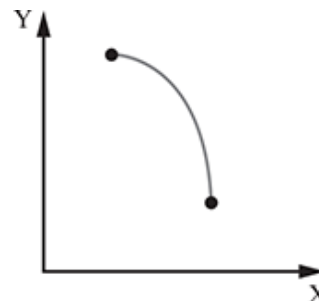
Linear positive slope



Linear negative slope

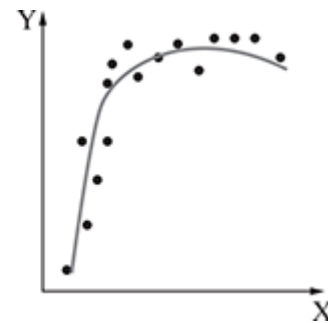
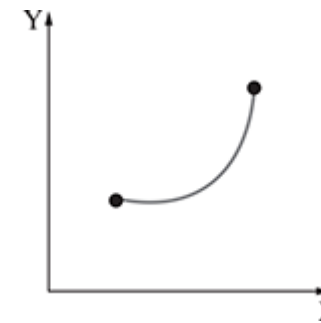
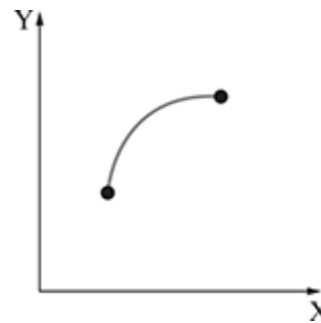
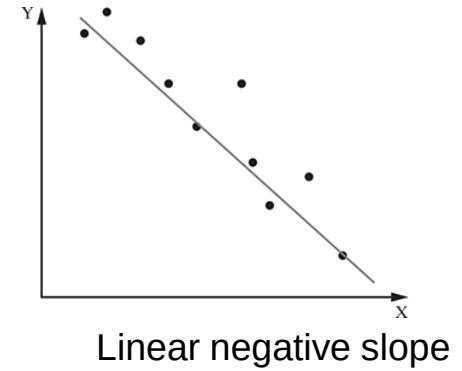
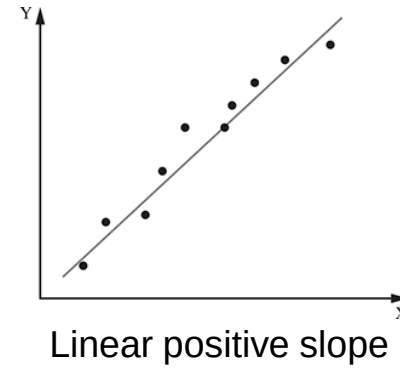
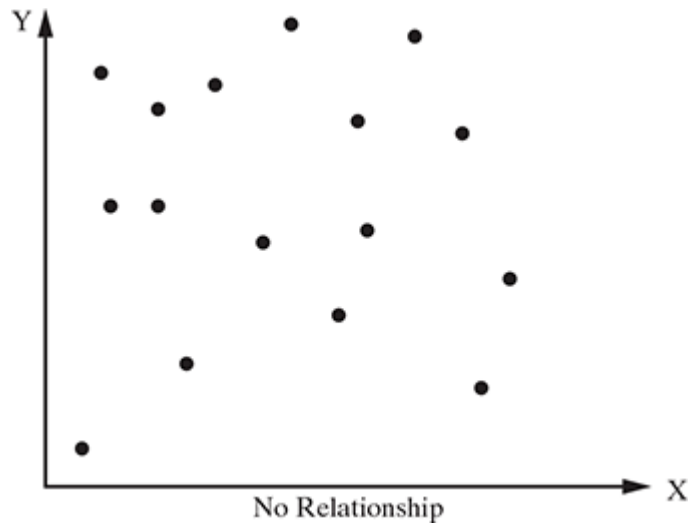


Curve linear positive slope

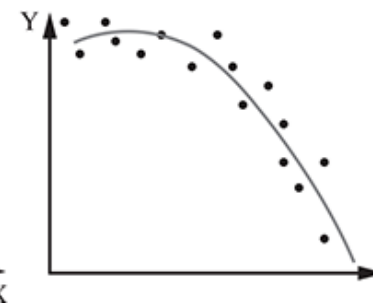
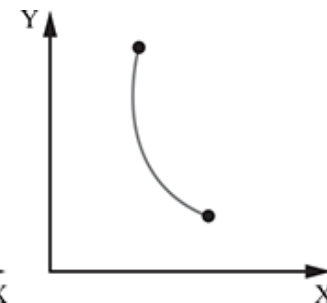
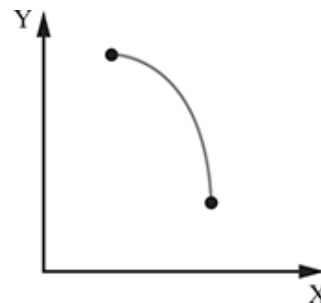


Curve linear negative slope

Relations in Regression



Curve linear positive slope



Curve linear negative slope

Simple Linear Regression: Least Square Method

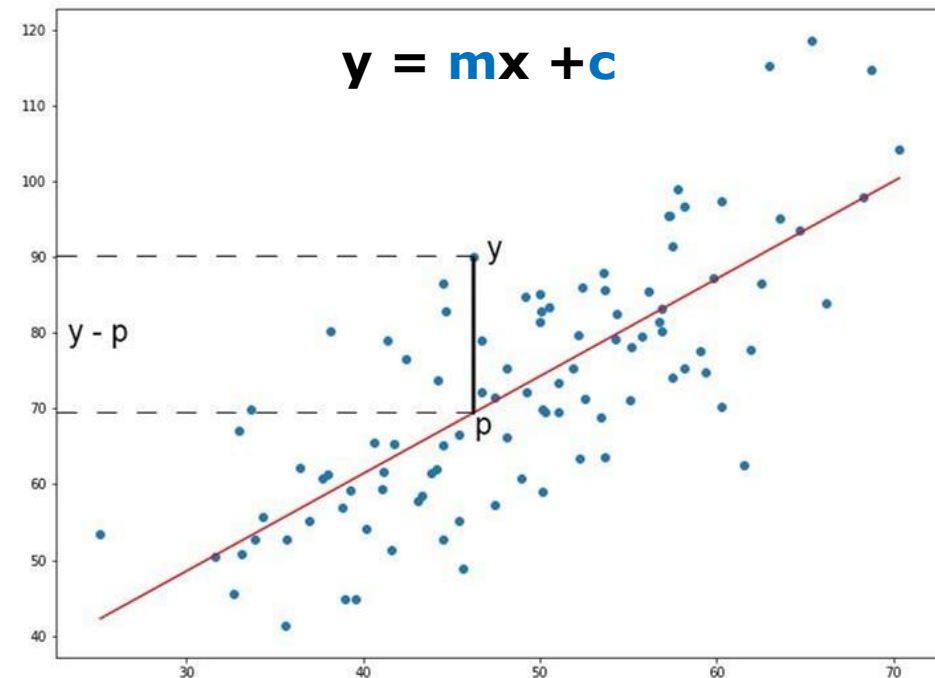
- How to find the best Regression Line?
- Our challenge is to determine the value of **m** and **c**, that gives the minimum error for the given dataset. We will be doing this by using the **Least Squares method**.
- **Loss function:**

$$L(x) = \sum_{i=1}^n (y_i - p_i)^2$$

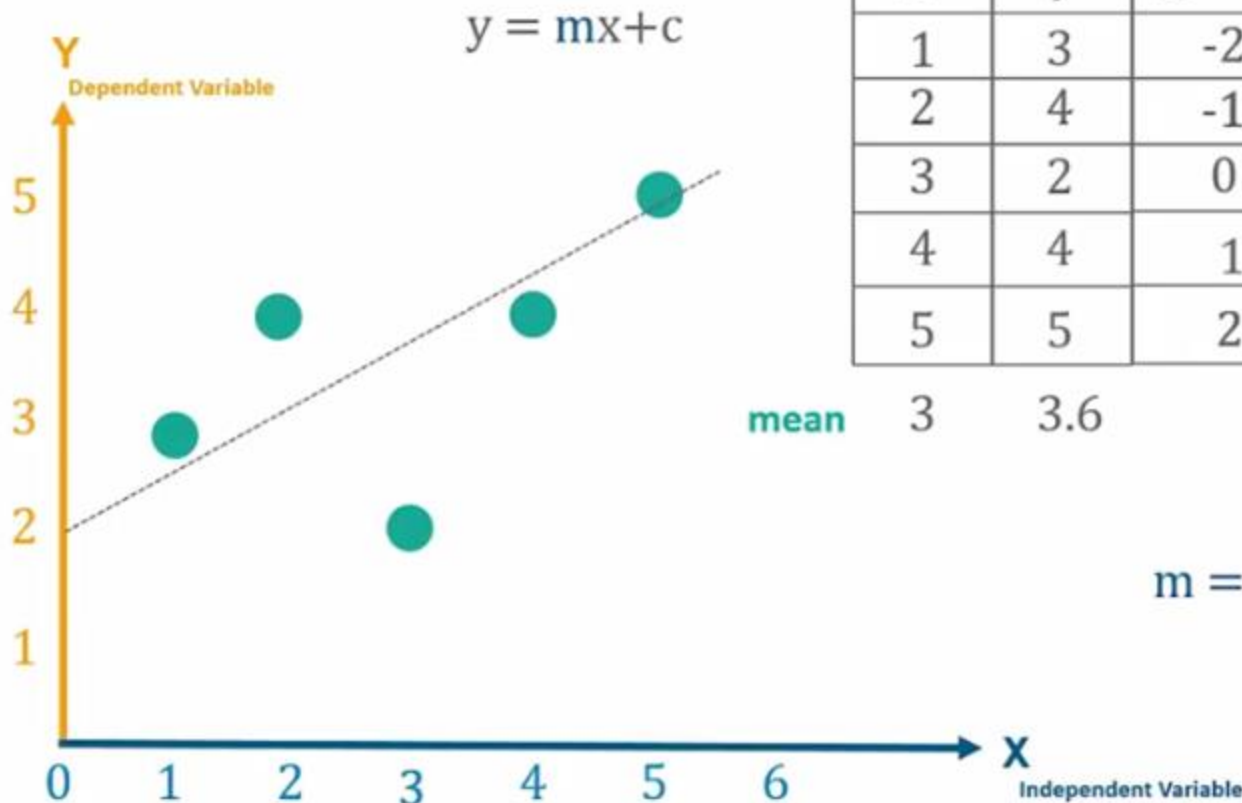
For minimum loss we take partial derivative of $L(x)$ and equate to 0, then finding expression of m and c .

$$m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$c = \bar{y} - m\bar{x}$$



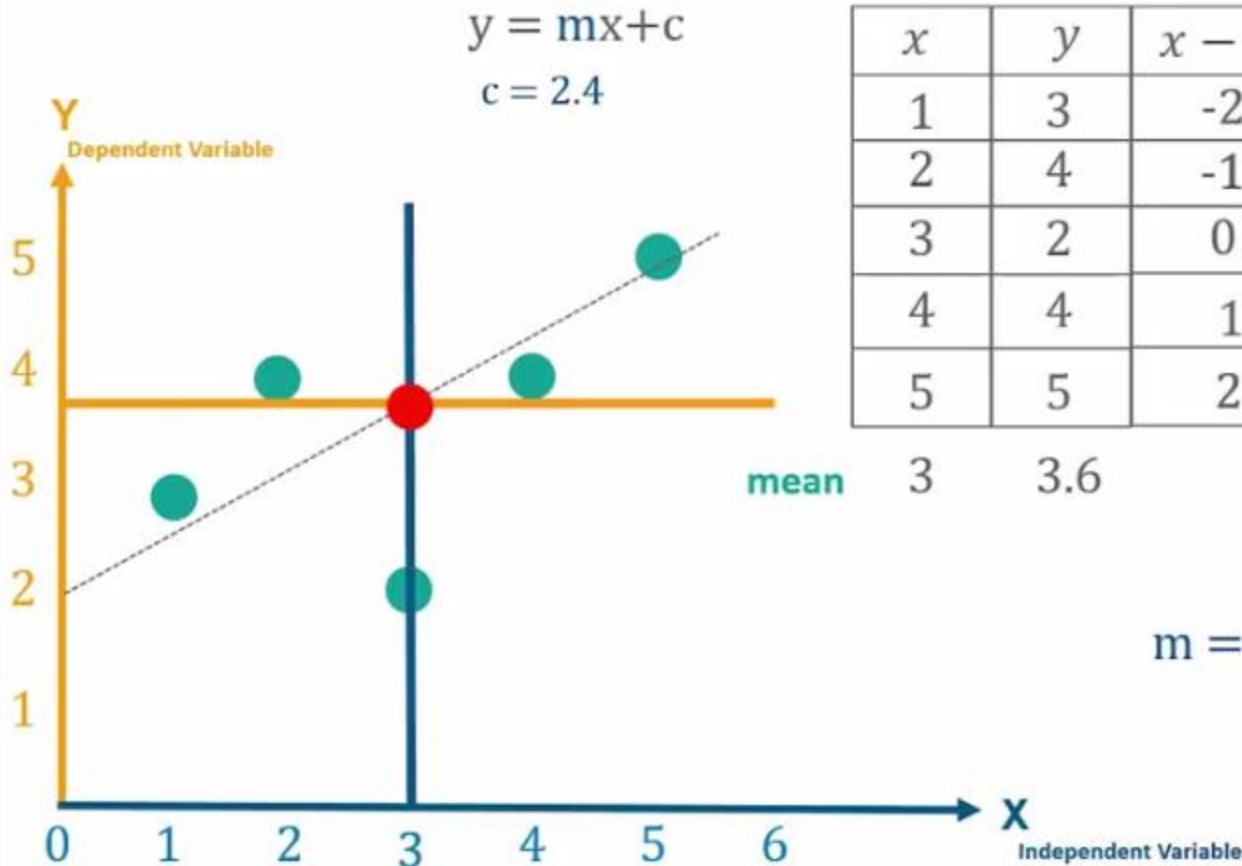
Simple Linear Regression: Least Square Method (Example)



x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$
1	3	-2	-0.6	4	1.2
2	4	-1	0.4	1	-0.4
3	2	0	-1.6	0	0
4	4	1	0.4	1	0.4
5	5	2	1.4	4	2.8
		mean 3	3.6	$\Sigma = 10$	$\Sigma = 4$

$$m = \frac{\Sigma (x - \bar{x})(y - \bar{y})}{\Sigma (x - \bar{x})^2} = \frac{4}{10}$$

Simple Linear Regression: Least Square Method (Example)



x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$
1	3	-2	-0.6	4	1.2
2	4	-1	0.4	1	-0.4
3	2	0	-1.6	0	0
4	4	1	0.4	1	0.4
5	5	2	1.4	4	2.8
3	3.6			$\Sigma = 10$	$\Sigma = 4$

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{4}{10}$$

$$m = 0.4$$

$$c = 2.4$$

$$y = 0.4x + 2.4$$

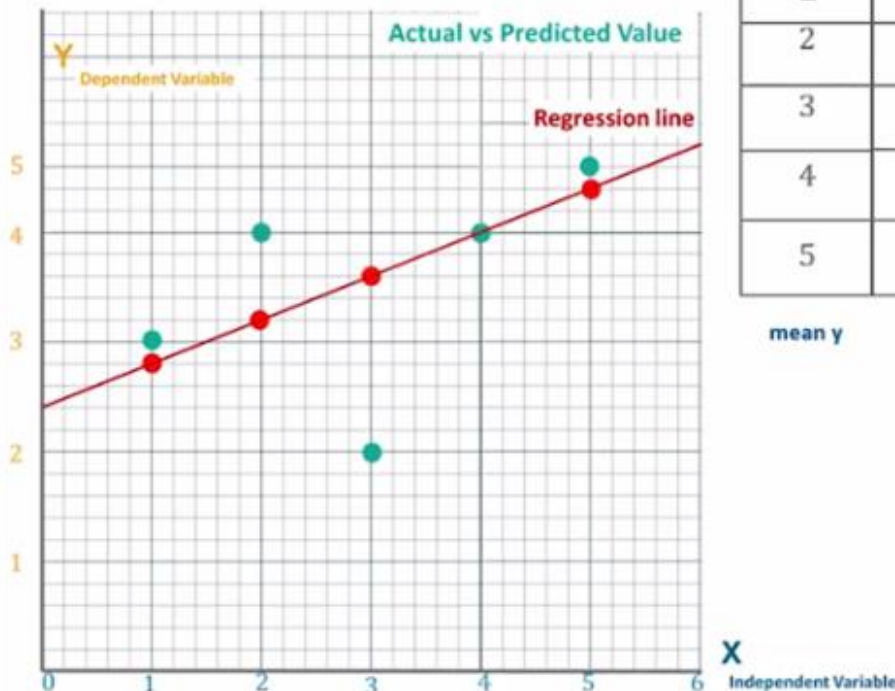
$$c = \bar{y} - m\bar{x}$$

Simple Linear Regression

- Measure of Goodness: R^2 method

$$R^2 = \frac{\sum (y_p - \bar{y})^2}{\sum (y - \bar{y})^2}$$

Calculation of R^2



x	y	$y - \bar{y}$	$(y - \bar{y})^2$	y_p	$(y_p - \bar{y})$	$(y_p - \bar{y})^2$
1	3	-0.6	0.36	2.8	-0.8	0.64
2	4	0.4	0.16	3.2	-0.4	0.16
3	2	-1.6	2.56	3.6	0	0
4	4	0.4	0.16	4.0	0.4	0.16
5	5	1.4	1.96	4.4	0.8	0.64
mean $\bar{y}$		3.6	$\sum$ 5.2		$\sum$ 1.6	

$$R^2 = \frac{1.6}{5.2} = \frac{\sum (y_p - \bar{y})^2}{\sum (y - \bar{y})^2}$$

OLS algorithm

- Step 1: Calculate the mean of X and Y
- Step 2: Calculate the errors of X and Y
- Step 3: Get the product
- Step 4: Get the summation of the products
- Step 5: Square the difference of X
- Step 6: Get the sum of the squared difference
- Step 7: Divide output of step 4 by output of step 6 to calculate 'b'
- Step 8: Calculate 'a' using the value of 'b'

Example of Simple Linear Regression

Internal Exam	15	23	18	23	24	22	22	19	19	16	24	11	24	16	23
External Exam	49	63	58	60	58	61	60	63	60	52	62	30	59	49	68

		Step 2		Step 3	Step 5
X	Y	X- mean (X)	Y- Mean (Y)	$(X_i - \bar{X})(Y_i - \bar{Y})$	$(X_i - \bar{X})^2$
15	49	-4.93	-7.8	38.454	24.3049
23	63	3.07	6.2	19.034	9.4249
18	58	-1.93	1.2	-2.316	3.7249
23	60	3.07	3.2	9.824	9.4249
24	58	4.07	1.2	4.884	16.5649
22	61	2.07	4.2	8.694	4.2849
22	60	2.07	3.2	6.624	4.2849
19	63	-0.93	6.2	-5.766	0.8649
19	60	-0.93	3.2	-2.976	0.8649
16	52	-3.93	-4.8	18.864	15.4449
24	62	4.07	5.2	21.164	16.5649
11	30	-8.93	-26.8	239.324	79.7449
24	59	4.07	2.2	8.954	16.5649
16	49	-3.93	-7.8	30.654	15.4449
23	68	3.07	11.2	34.384	9.4249
19.9	56.8		$\Sigma(X_i - \bar{X})(Y_i - \bar{Y})$	429.8	226.9335

Step 1

Step 4

Step 6

Step 7: Divide (step4 / step6)

$$b = 429.28 / 226.93 = 1.89$$

Step 8: Calculate a using the value of b

$$a = \bar{Y} - b\bar{X}$$

$$a = 56.8 - 1.89 \times 19.9$$

$$a = 19.05$$

Calculation summary:

Sum of X = 299

Sum of Y = 852

Mean X, M = 19.93

Mean Y, M = 56.8

$$b = \frac{\Sigma_i(X_i - \bar{X})(Y_i - \bar{Y})}{\Sigma_i(X_i - \bar{X})^2} = \frac{\text{cov}(X, Y)}{\text{Var}(X)}$$

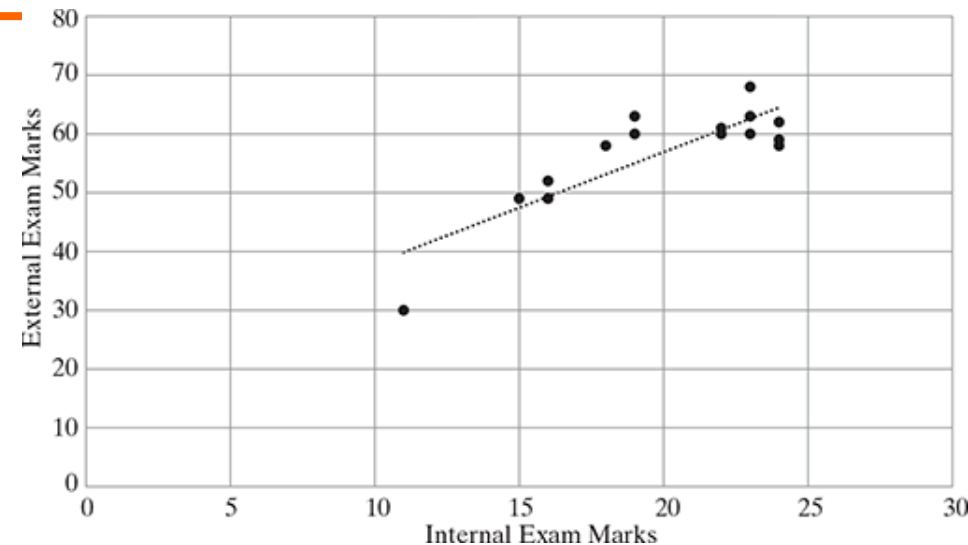
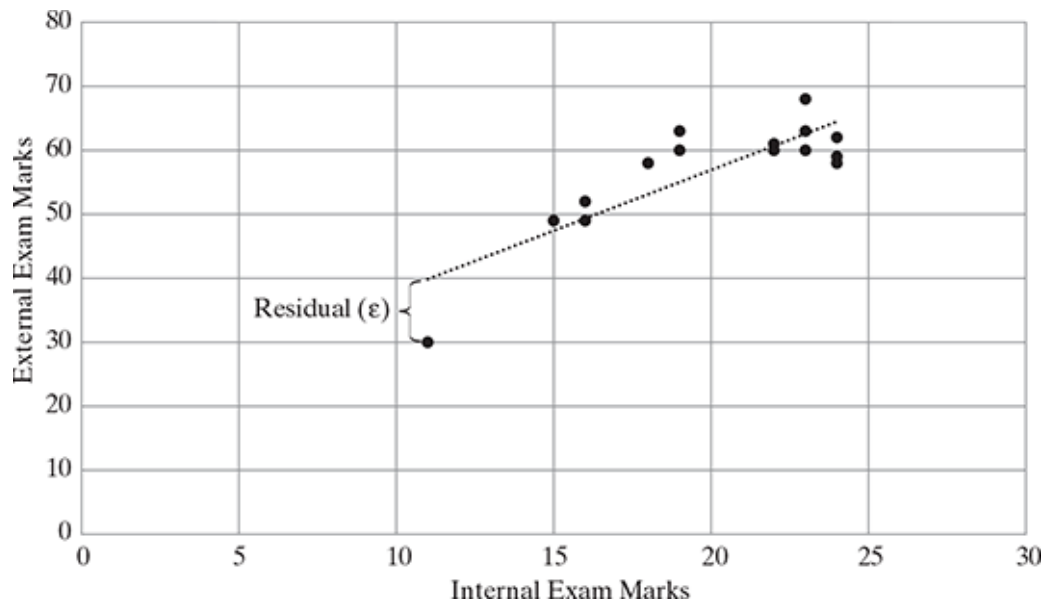
$$a = \bar{Y} - b\bar{X}$$

Error in Simple Regression

$$Y = (a + bX) + \epsilon$$

Example of simple regression

Internal Exam	15	23	18	23	24	22	22	19	19	16	24	11	24	16	23
External Exam	49	63	58	60	58	61	60	63	60	52	62	30	59	49	68



Scatter plot and regression line

Sum of Square of Residual

$$SSE = \left(\text{i.e. } \sum_i \epsilon_i^2 \right).$$

Residual is the distance between the predicted point (on the regression line) and the actual point as depicted in Figure

Multiple Linear Regression

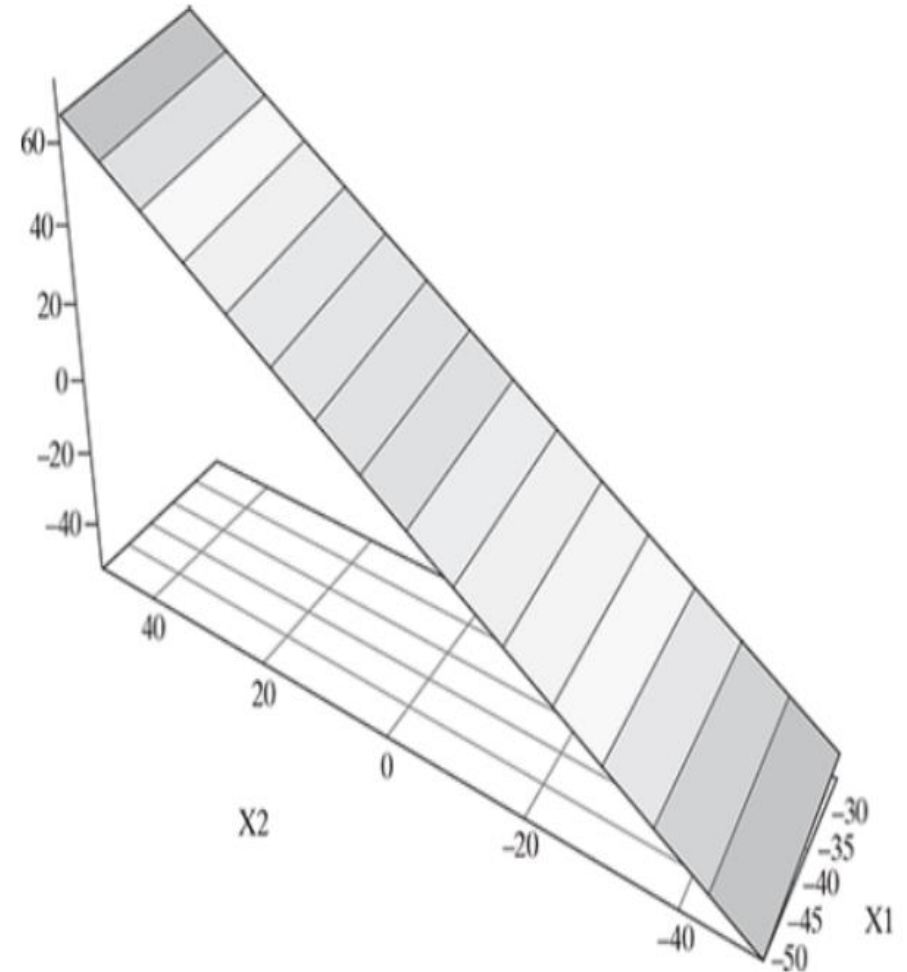
- **Two or more independent variables, i.e. predictors** are involved in the model.
- In the example of simple linear regression by considering Price of a Property as the dependent variable and the Area of the Property (in sq. m.) as the predictor variable.
- If we consider Price of a Property (in \$) as the dependent variable and Area of the Property (in sq. m.), location, floor, number of years since purchase and amenities available as the independent variables, we can form a **multiple regression equation** as shown below:

$$\text{Price}_{\text{Property}} = f(\text{Area}_{\text{Property}}, \text{location}, \text{floor}, \text{Ageing}, \text{Amenities})$$

- The simple linear regression model and the multiple regression model assume that the **dependent variable is continuous**.
- The following expression describes the equation involving the relationship with two predictor variables, namely X_1 and X_2 .
$$\hat{Y} = a + b_1X_1 + b_2X_2$$
- The model describes a plane in the **three-dimensional space of $\hat{Y}$, X_1 , and X_2** .
- Parameter '**a**' is the **intercept** of this plane. Parameters ' b_1 ' and ' b_2 ' are referred to as **partial regression coefficients**.
- **Parameter b_1 represents** the change in the mean response corresponding to a unit change in X_1 when X_2 is held constant.
- **Parameter b_2 represents** the change in the mean response corresponding to a unit change in X_2 when X_1 is held constant.

- Consider the following example of a multiple linear regression model with two predictor variables, namely X_1 and X_2 .

$$\hat{Y} = 22 + 0.3X_1 + 1.2X_2$$



Multiple regression for estimating equation when there are 'n' predictor variables is as follows:

$$\hat{Y} = a + b_1X_1 + b_2X_2 + b_3X_3 + \dots + b_nX_n$$

While finding the best fit line, we can fit either a polynomial or curvilinear regression. These are known as **polynomial or curvilinear regression, respectively.**

Assumptions in Regression Analysis

1. The dependent variable (Y) can be calculated / predicated as a linear function of a specific set of independent variables (X's) plus an error term (ϵ).
2. The number of observations (n) is greater than the number of parameters (k) to be estimated, i.e. $n > k$.
3. Relationships determined by regression are **only relationships of association based on the data set** and not necessarily of cause and effect of the defined class.
4. Regression line can be **valid only over a limited range of data**. If the line is extended (outside the range of extrapolation), it may only lead to wrong predictions.

5. If the business conditions change and the business assumptions underlying the regression model are no longer valid, then the past data set will no longer be able to predict future trends.
6. Variance is the same for all values of X (**homoskedasticity**).
7. The error term (ϵ) is normally distributed. This also means that the mean of the error (ϵ) has an expected value of 0.
8. The values of the error (ϵ) are independent and are not related to any values of X. This means that there are no relationships between a particular X, Y that are related to another specific value of X, Y.

Given the above assumptions, the OLS estimator is the Best Linear Unbiased Estimator (BLUE), and this is called as Gauss-Markov Theorem.

Main Problems in Regression Analysis

- Two primary problems: **Multicollinearity and heteroskedasticity**

Multicollinearity

- Two variables are perfectly **collinear** if there is an exact linear relationship between them.
- Multicollinearity is the situation in which the degree of correlation is not only between the dependent variable and the independent variable, but there is also a **strong correlation within (among) the independent variables themselves.**
- A multiple regression equation can make good predictions when there is multicollinearity, but it is difficult for us to determine how the dependent variable will change if each independent variable is changed one at a time.

- When multicollinearity is present, it increases the standard errors of the coefficients.
- One way to gauge multicollinearity is to calculate the **Variance Inflation Factor (VIF)**, which assesses how much the variance of an estimated regression coefficient increases if the predictors are correlated.
- If no factors are correlated, the VIFs will be equal to 1.
- The assumption of **no perfect collinearity** states that there is no exact linear relationship among the independent variables.
- This assumption implies two aspects of the data on the independent variables.

- First, none of the independent variables, other than the variable associated with the intercept term, can be a constant.
- Second, variation in the X's is necessary.
- In general, **the more variation in the independent variables, the better will be the OLS** estimates in terms of identifying the impacts of the different independent variables on the dependent variable.

Heteroskedasticity

- Refers to the changing variance of the error term.
- If the variance of the error term is not constant across data sets, there will be erroneous predictions.
- In general, for a regression equation to make accurate predictions, the error term should be independent, identically (normally) distributed (iid).

Mathematically, this assumption is written as

$$\begin{aligned}\text{var}(u_i|X) &= \sigma^2 \quad \text{and} \\ \text{cov}(u_i u_j|X) &= 0 \quad \text{for } i \neq j\end{aligned}$$

where ‘var’ represents the variance, ‘cov’ represents the covariance, ‘ u ’ represents the error terms, and ‘ X ’ represents the independent variables.

This assumption is more commonly written as

$$\begin{aligned}\text{var}(u_i) &= \sigma^2 \quad \text{and} \\ \text{cov}(u_i u_j) &= 0 \quad \text{for } i \neq j.\end{aligned}$$

Improving Accuracy of the Linear Regression Model

- **Accuracy** refers to how close the estimation is near the actual value, whereas **prediction** refers to continuous estimation of the value.

High bias = low accuracy (not close to real value)

High variance = low prediction (values are scattered)

Low bias = high accuracy (close to real value)

Low variance = high prediction (values are close to each other)

- We have a regression model which is **highly accurate and highly predictive**; therefore, the overall error of our model will be low, implying a low bias (high accuracy) and low variance (high prediction). **This is highly preferable.**

Improving Accuracy of the Linear Regression Model

Accuracy of linear regression can be improved using the following three methods:

1. Shrinkage Approach
2. Subset Selection
3. Dimensionality (Variable) Reduction

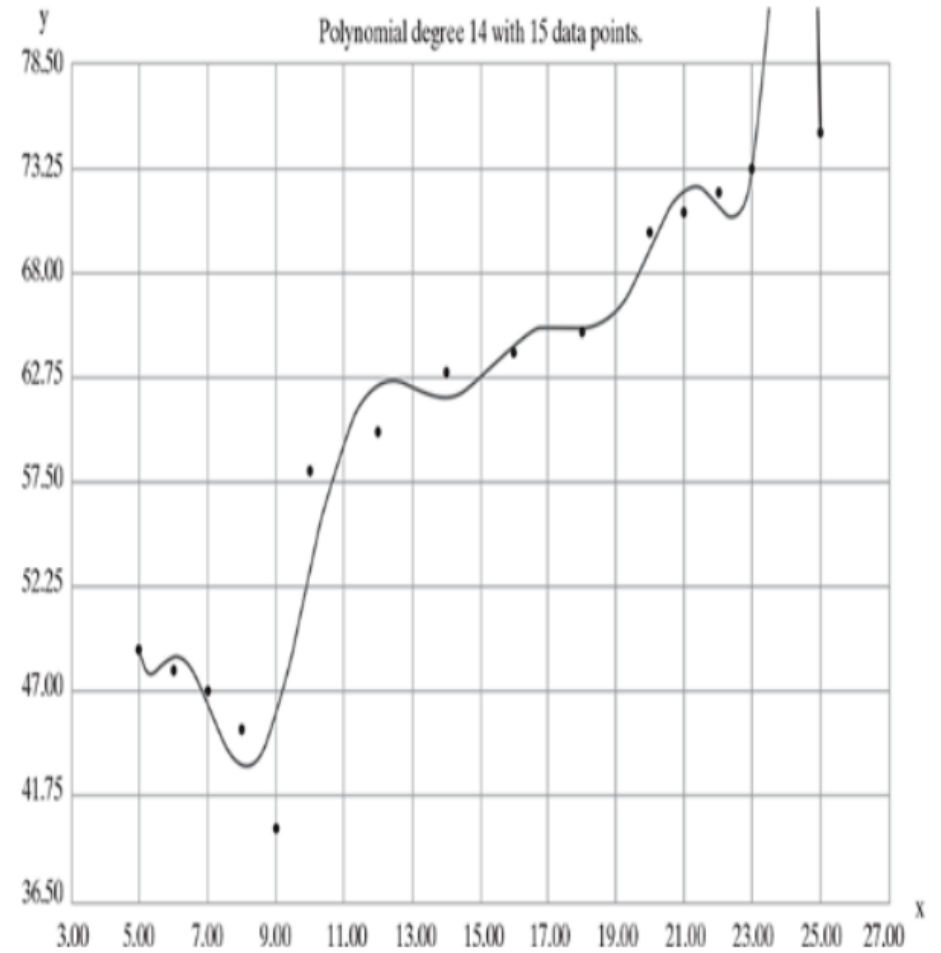
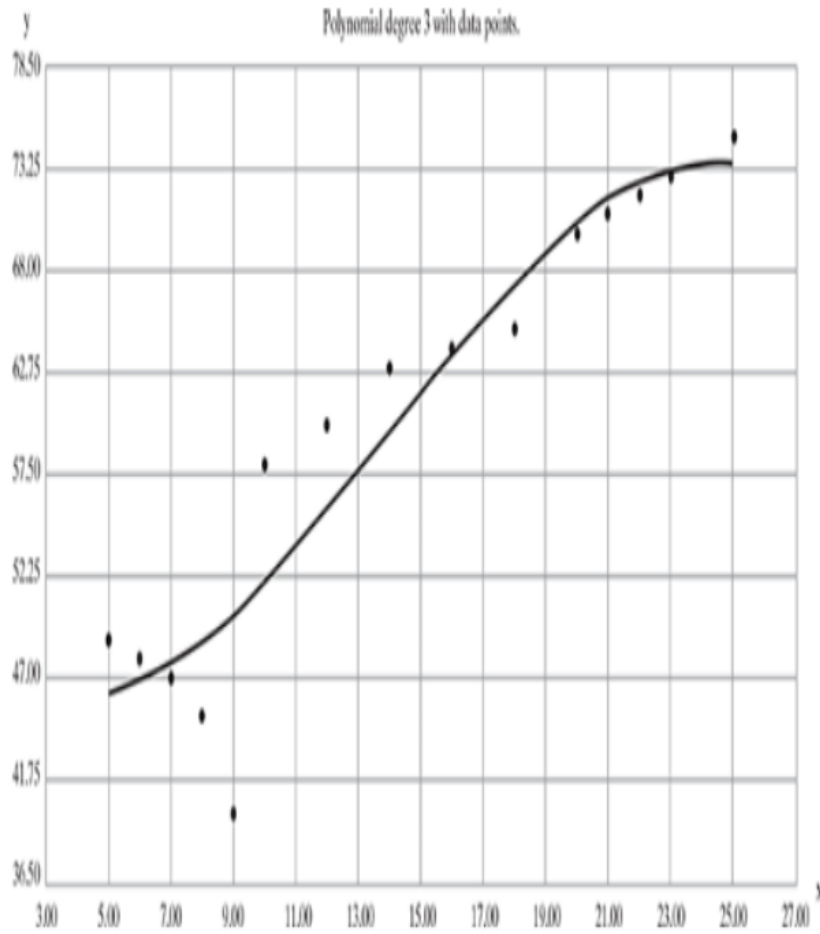
Polynomial Regression Model

- Extension of the simple linear model by adding extra predictors obtained by raising (squaring) each of the original predictors to a power.
- This approach provides a simple way to yield a non-linear fit to data. For example

$$f(x) = c_0 + c_1.X^1 + c_2.X^2 + c_3.X^3$$

- Let us use the below data set of (X, Y) for degree 3 polynomial.

Internal Exam (X)	15	23	18	23	24	22	22	19	19	16	24	11	24	16	23
External Exam (Y)	49	63	58	60	58	61	60	63	60	52	62	30	59	49	68



- As you can observe, the regression line is slightly curved for polynomial degree 3 with the above 15 data points.
- **The regression line will curve further** if we increase the polynomial degree.
- At the extreme value as shown above, the regression line will be overfitting into all the original values of X.

What is Logistic Regression?

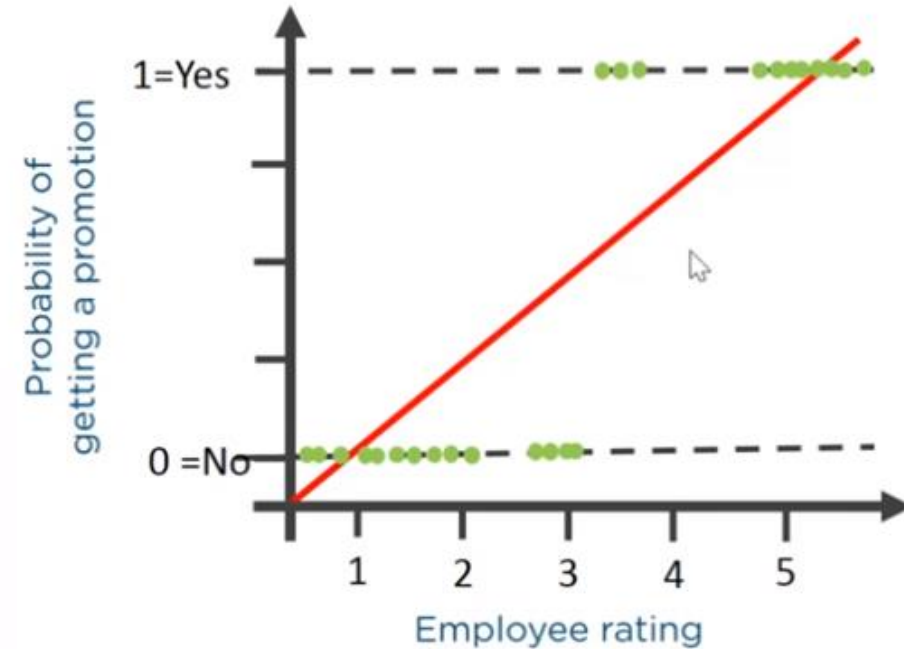
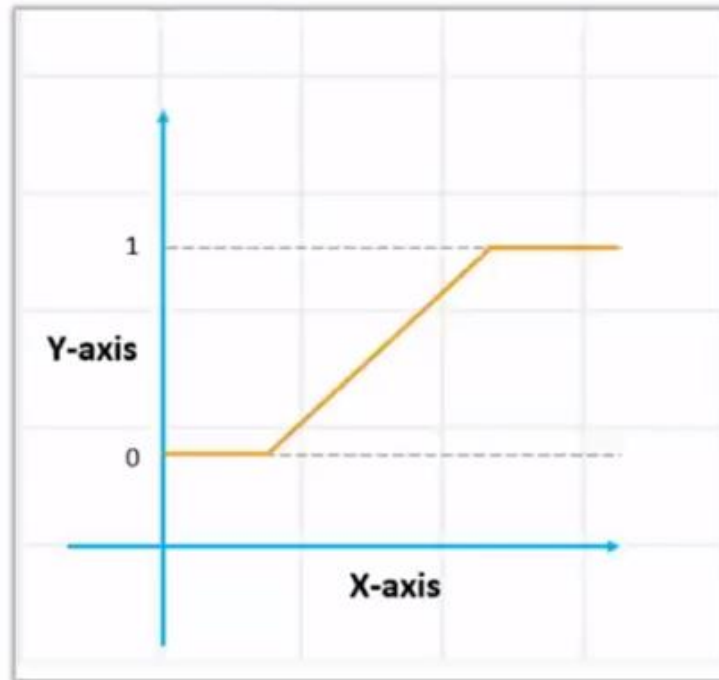
- Logistic regression is a **Classification algorithm**.
- **Logistic Regression is all about predicting binary variables, not predicting continuous variables.**
- Logistic regression models estimate how probability of an event may be affected by one or more explanatory variables.
- Logistic regression is a technique used for predicting “**class probability**”, that is the probability that the case belongs to a particular class.

Use cases of Logistic Regression

- Mail [Spam / Not Spam]
- Transaction [Fraudulent / Normal]
- Tumor [Malignant / Benign]
- Sentimental Analysis [Positive / Negative]
- Weather Prediction [Rain / Not Rain]
- Medical Diagnosis [Fit / ill]

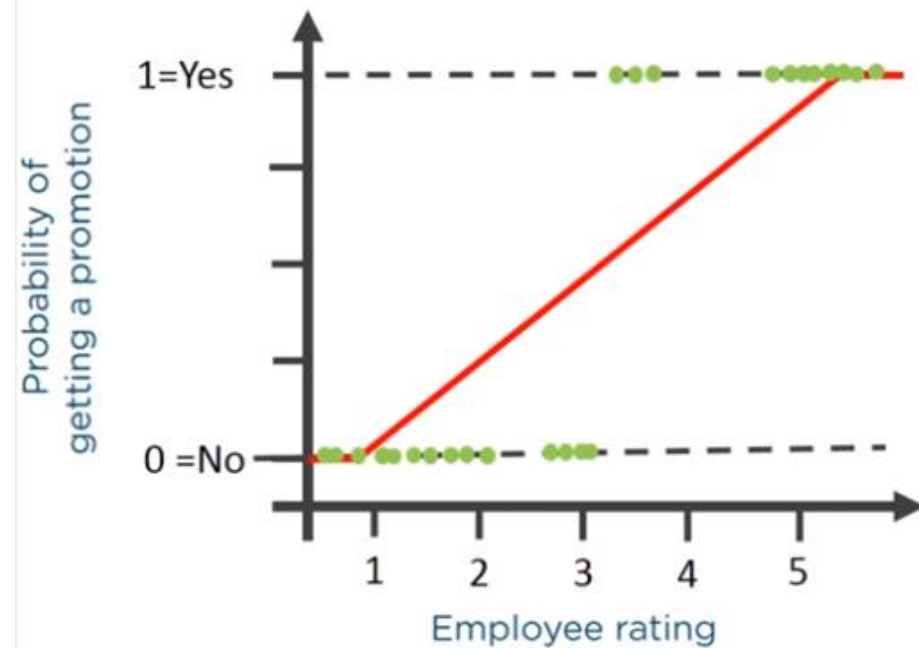
Linear and Logistic Regression

What if you wanted to know whether the employee would get a promotion or not based on their rating

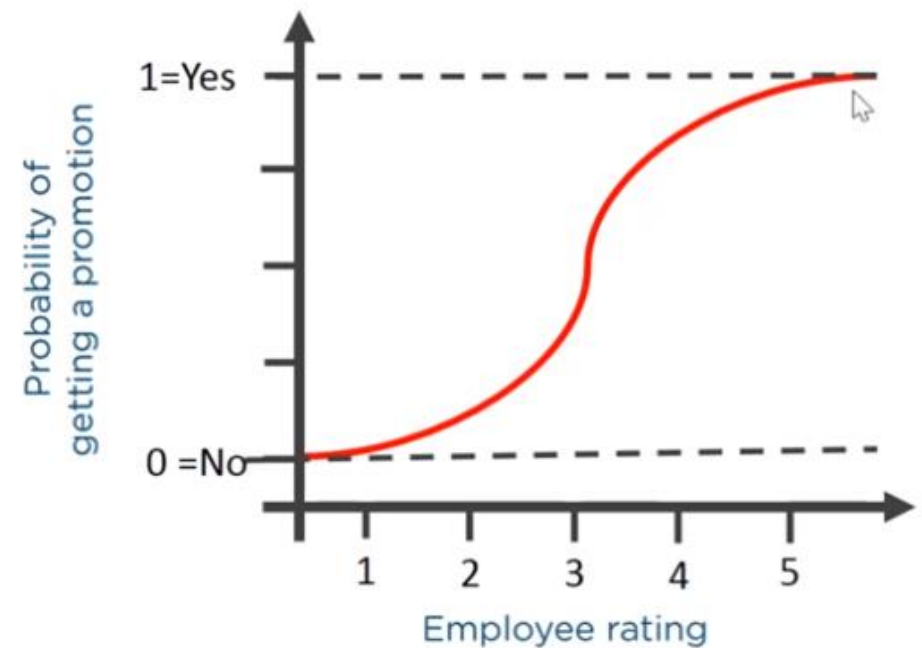


Linear and Logistic Regression

Logistic curve
Sigmoid (S) curve

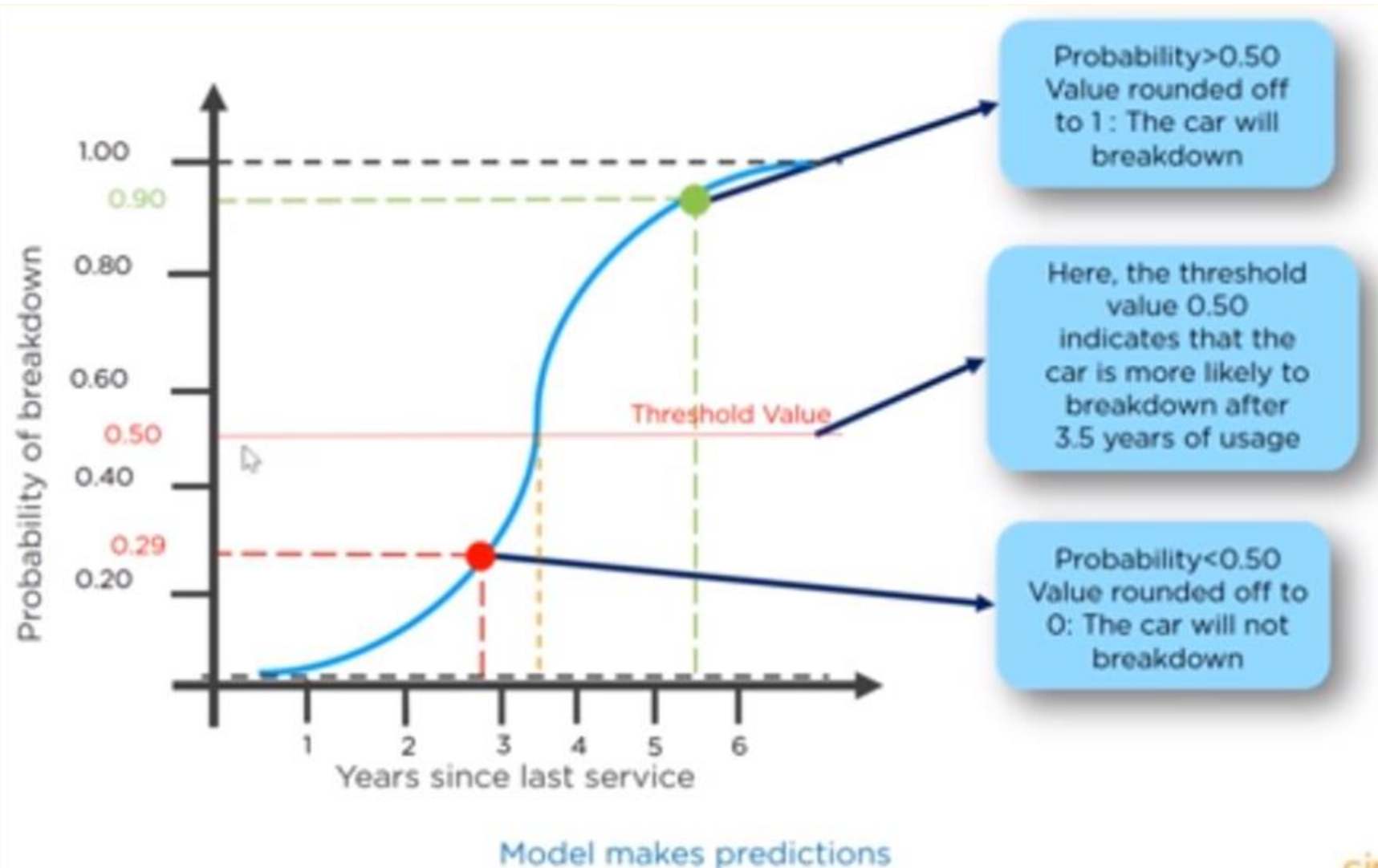


So, how did this...



...become this?

Logistic Regression Curve



Some fundamentals terms of Logistic Regression

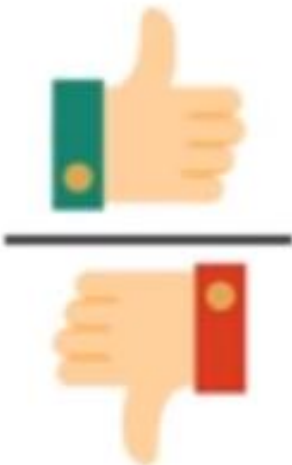
- The **probability** that an event will occur is the fraction of times you expect to see that event in many trials. If the probability of an event occurring is Y , then the probability of the event not occurring is $1 - Y$. **Probabilities** always range between 0 and 1.
- The **odds** are defined as the probability that the event will occur divided by the probability that the event will not occur. Unlike **probability**, the odds are not constrained to lie between 0 and 1 but can take any value from zero to infinity.
- If the probability of Success is P , then the odds of that event is:

$$odds = \frac{P}{1 - P}$$

- The **logit function** is the logarithmic transformation of the logistic function. It is defined as the natural logarithm of odds.

Math behind Logistic Regression

To understand Logistic Regression, let's talk about the odds of success

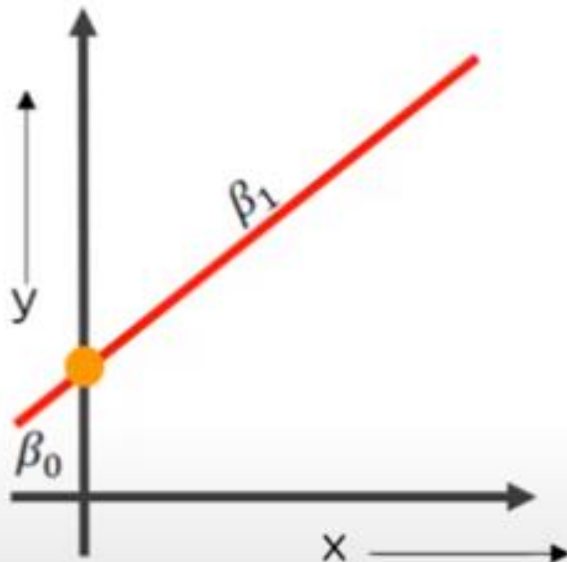
Odds (θ) = 

Probability of an event happening
Probability of an event not happening *or* $\theta = \frac{p}{1-p}$

The values of odds range from 0 to ∞
The values of probability change from 0 to 1

Math behind Logistic Regression

Take the equation of the straight line



Here, β_0 is the y-intercept
 β_1 is the slope of the line
 x is the value of the x co-ordinate
 y is the value of the prediction

The equation would be: $y = \beta_0 + \beta_1 x$

Math behind Logistic Regression

Now, we predict the odds of success

$$\log \left(\frac{p(x)}{1-p(x)} \right) = \beta_0 + \beta_1 x$$

Exponentiating both sides:

$$e^{\ln \left(\frac{p(x)}{1-p(x)} \right)} = e^{\beta_0 + \beta_1 x}$$

$$\left(\frac{p(x)}{1-p(x)} \right) = e^{\beta_0 + \beta_1 x}$$

Let $Y = e^{\beta_0 + \beta_1 x}$

Then $\frac{p(x)}{1-p(x)} = Y$

$$p(x) = Y(1 - p(x))$$

$$p(x) = Y - Y(p(x))$$

$$p(x) + Y(p(x)) = Y$$

$$p(x)(1 + Y) = Y$$

$$p(x) = \frac{Y}{1 + Y}$$

$$p(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

The equation of a sigmoid function:

$$p(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

- Let us say we have a model that can predict whether a person is male or female on the basis of their height.
- Given a height of 150 cm, we need to predict whether the person is male or female.
- We know that the coefficients of $a = -100$ and $b = 0.6$.
- Using the above equation, we can calculate the probability of male given a height of 150 cm or more formally $P(\text{male}|\text{height} = 150)$.

$$y = e^{(a + b \times X)} / (1 + e^{(a + b \times X)})$$
$$y = \exp(-100 + 0.6 \times 150) / (1 + \text{EXP}(-100 + 0.6 \times X))$$
$$y = 0.000046$$

or a probability of near zero that the person is a male.

Linear vs Logistic Regression

Basis	Linear Regression	Logistic Regression
Core concept (Modeling of data)	Data is modeled using a straight line.	Data is modeled using a logistic (sigmoid) function.
Used with	Continuous variable	Categorical variable
Output/prediction	Value of the variable	Probability of occurrence of event
Problem Solved	Regression	Classification
Accuracy (goodness of fit)	Loss, R^2 , adjusted R^2 , etc.	Accuracy, Precision, Recall, F1 score, ROC curve, Confusion matrix, etc.

- The basic difference: the type of function that is used for mapping
 - (Linear: continuous $X \rightarrow$ Continuous Y ;
 - Logistic: continuous $X \rightarrow$ binary Y) – used for deciding category or true/false decisions of the data

Parameter Estimation by Maximum Likelihood Method

- The coefficients in a logistic regression are estimated using a process called Maximum Likelihood Estimation (MLE).

- Likelihood function:

$$L(\theta|x) = \prod_{i=1}^n f(x_i|\theta)$$

MLE is about predicting the value for the parameters that maximizes the likelihood function.

$$\log L(\theta|x) = \sum_{i=1}^n \log f(x_i|\theta)$$

When we use the Bernoulli distribution represents each flip of the coin:

$$f(x_i|\theta) = \theta^{x_i}(1 - \theta)^{1-x_i}$$

Each observation X_i is independent and also identically distributed (iid), and the joint distribution simplifies to a product of distributions.

$$f(x_1, \dots, x_n|\theta) = \prod_{i=1}^n f(x_i|\theta) = \theta^{x_1}(1 - \theta)^{1-x_1} \dots \theta^{x_n}(1 - \theta)^{1-x_n} = \theta^{\sum x_i} (1 - \theta)^{n - \sum x_i},$$

where $\sum x_i$ is the number of flips that resulted in the expected outcome (heads in this case).

Thank You

Parameter Estimation by Maximum Likelihood Method

- Probability density function for binary logistic regression is given by:

$$P(y_i) = p_i^{y_i} (1 - p_i)^{1-y_i}$$

$$L(\beta) = P(y_1, y_2, \dots, y_n) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}$$

$$\ln(L(\beta)) = \sum_{i=1}^n y_i \ln[p(x_i)] + \sum_{i=1}^n (1 - y_i) [\ln(1 - p_i(x_i))]$$

$$\ln[L(\beta)] = \sum_{i=1}^n y_i (\beta_0 + \beta_1 x_i) - \sum_{i=1}^n \ln(1 + \exp(\beta_0 + \beta_1 x_i))$$

Parameter Estimation by Maximum Likelihood Method

$$\frac{\partial \ln(L(\beta_0, \beta_1))}{\partial \beta_0} = \sum_{i=1}^n y_i - \sum_{i=1}^n \frac{\exp(\beta_0 + \beta_1 x_i)}{1 + \exp(\beta_0 + \beta_1 x_i)} = 0$$

$$\frac{\partial \ln(L(\beta_0, \beta_1))}{\partial \beta_1} = \sum_{i=1}^n x_i y_i - \sum_{i=1}^n \frac{x_i \exp(\beta_0 + \beta_1 x_i)}{1 + \exp(\beta_0 + \beta_1 x_i)} = 0$$

The above system of equations are solved iteratively to estimate β_0 and β_1

References

- Coursera tutorial - Linear Regression, Logistic Regression
- SimpliLearn tutorial – Logistic Regression
- Wikipedia – linear regression
- Business Analytics – by U.Dinesh Kumar